# gpeyt1-11 · OSHA sloping

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

## Problem

A 13 ft deep utility trench with a 5 ft wide bottom will be sloped (no shoring) on both sides through a medium-stiff silty clay classified by the competent person as OSHA Type B (qu = 1.0 tsf, fissured, previously disturbed). The trench run is 240 ft long. The spoil pile will be kept 2 ft back from the edge, and the excavator weighs 48,000 lb.

**Most nearly, the minimum top width of the trench is:**

A. 24 ft
B. 44 ft
C. 31 ft
D. 18 ft

## Given equations / data

> OSHA maximum allowable slopes (H:V): Type A 3/4:1, Type B 1:1, Type C 1½:1 (excavations ≤ 20 ft)

## Worked solution

1. Type B → 1H:1V on each side
2. Horizontal run each side = 13 × 1 = 13 ft
3. Top width = 5 + 2(13) = 31.0 ft
4. (Excavated volume ≈ (5+31.0)/2 × 13 × 240/27 = 2,080 yd³ — not asked)

## Final answer

**31 ft**

## Why the other values are wrong

- **24 ft** — used the Type A slope 3/4:1
- **44 ft** — used the Type C slope 1½:1
- **18 ft** — sloped only one side

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.10.2 Slope and Shield Configurations (Allowable Slopes table), p. 152; 29 CFR 1926 Subpart P App. B

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=hE7KZATybxA, https://www.youtube.com/watch?v=5IweGa4yOBc (no creator values copied).

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# gpeyt1-12 · Sand-cone relative compaction

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## Problem

A field density test on a compacted embankment lift: the mass of calibrated sand that filled the hole (after subtracting the cone correction) was 3.36 lb; the bulk density of the calibrated sand is 93.5 pcf. The moist soil removed from the hole weighed 4.66 lb, and its oven-dry water content is 11.8%. The lab standard Proctor gave $\gamma_{d,max}$ = 121.0 pcf at $w_{opt}$ = 10.5%. Specification: ≥ 95% relative compaction, w within −1/+3% of optimum. Lift thickness 8 in.

**Most nearly, the relative compaction and the test result are:**

A. 107.2%, passes
B. 94.5%, fails
C. 95.9%, passes
D. 104.3%, passes

## Given equations / data

> $V = W_{sand}/\gamma_{sand}$; $\gamma_t = W/V$; $\gamma_d = \gamma_t/(1 + w)$; $RC = \gamma_d/\gamma_{d,max} \times 100\%$

## Worked solution

1. Hole volume $V = 3.36/93.5 = 0.0359$ ft³
2. $\gamma_t = 4.66/0.0359 = 129.7$ pcf
3. $\gamma_d = 129.7/1.118 = 116.0$ pcf
4. $RC = 116.0/121.0 = 95.9\%$ (≥ 95% required); w = 11.8% is +1.3% of opt (within −1/+3) → passes

## Final answer

**95.9%, passes**

## Why the other values are wrong

- **107.2%, passes** — used moist unit weight (forgot ÷(1+w))
- **94.5%, fails** — multiplied by (1 − w) instead of dividing by (1 + w)
- **104.3%, passes** — inverted the ratio

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.9.1–3.9.2 Laboratory and Field Compaction, pp. 146–148; §3.8.3 Weight-Volume Relationships

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# gpeyt1-13 · Compaction production

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## Problem

A contractor will compact 18,000 compacted yd³ of embankment with a single smooth-drum vibratory roller. Drum compacted width per pass = 7.0 ft; average speed = 2.5 mph; compacted lift = 8 in.; job efficiency = 80% (48 min/hr); test strip shows 6 passes are needed for 95% RC. Shrinkage from bank to compacted is 15% and the haul fleet delivers 300 bank yd³/hr.

**Most nearly, how many roller-hours are required?**

A. 10 hr
B. 47 hr
C. 59 hr
D. 50 hr

## Given equations / data

> Compacted yd³/hr = 16.3 · W · S · L · efficiency / n (W ft, S mph, L in.)

## Worked solution

1. Rate = 16.3 × 7.0 × 2.5 × 8 × 0.8 / 6 = 304.3 compacted yd³/hr
2. Hours = 18,000/304.3 = 59 hr
3. Shrinkage only converts compacted to bank yd³ — not needed because quantity is already compacted

## Final answer

**59 hr**

## Why the other values are wrong

- **10 hr** — multiplied by n instead of dividing
- **47 hr** — omitted the efficiency factor
- **50 hr** — applied the shrinkage factor to compacted quantity

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §2.3.3.1 Production Rate for Soil Compaction (Ch. 2 Construction)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc (no creator values copied).

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# gpeyt1-14 · Moisture conditioning

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## Problem

Borrow for a 52,000 ft³ (compacted) highway embankment has an in-situ water content of 7.5%. The specification requires placement at 13.0% and a dry unit weight of 114 pcf. The water truck holds 4,000 gal. The borrow's in-place moist unit weight is 118 pcf and Gs = 2.68 (not needed).

**Most nearly, the volume of water that must be added is:**

A. 39,140 gal
B. 92,514 gal
C. 5,225 gal
D. 40,514 gal

## Given equations / data

> Gallons = γd,desired × Vcompacted × (wdesired − wborrow)/100 / 8.33 lb/gal

## Worked solution

1. Δw = 13.0 − 7.5 = 5.5%
2. Water weight = 114 × 52,000 × 0.055 = 326,040 lb
3. Gallons = /8.33 = 39,140 gal (≈ 10 truckloads)

## Final answer

**39,140 gal**

## Why the other values are wrong

- **92,514 gal** — used the full desired w (ignored borrow moisture)
- **5,225 gal** — divided by 62.4 (got ft³, labeled gal)
- **40,514 gal** — used the moist borrow unit weight

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §2.3.1 Earthwork/compaction — amount of water to add (Ch. 2 Construction)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v_1tQfxBowQ, https://www.youtube.com/watch?v=f02U9lj1I58 (no creator values copied).

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# gpeyt1-15 · Piezometer QA

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## Problem

A vibrating-wire piezometer is installed at the middle of a soft clay layer (tip at El. 60.0) beneath a staged embankment. Immediately after stage-1 fill, the excess pore pressure at the tip was 1,150 psf. Three months later the piezometer indicates a total head equivalent to a water level at El. 96.5, while the static groundwater table is at El. 88.0. Settlement plates show 7.2 in. of settlement; inclinometers show <0.3 in. lateral movement.

**Most nearly, the local degree of consolidation at the piezometer tip is:**

A. 54%
B. 46%
C. 76%
D. 19%

## Given equations / data

> ue = γw × (piezometric elevation − static water elevation); Uz = 1 − ue/ue0; γw = 62.4 pcf

## Worked solution

1. Excess head = 96.5 − 88.0 = 8.5 ft
2. ue = 8.5 × 62.4 = 530 psf
3. Uz = 1 − 530/1,150 = 0.539 ≈ 54%

## Final answer

**54%**

## Why the other values are wrong

- **46%** — reported ue/ue0 (fraction still remaining)
- **76%** — measured excess head from the ground surface (El. 92.0) instead of the static water table
- **19%** — used the clay unit weight (110 pcf) instead of γw

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.11 Geotechnical Instrumentation, p. 156; §3.2.3 consolidation

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